Even Harmonics in Three-Phase Active Front-End Rectifiers: Origins, Mechanisms, and Mitigation

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Abstract

Even-order current harmonics are non-characteristic for a three-phase active front-end (AFE) rectifier: under ideal half-wave symmetry they cannot exist. Their appearance in the field is therefore always a symptom of a symmetry-breaking mechanism in the measurement chain, in the power stage, in the modulator, or in the grid itself. This note reviews the physical and control-theoretic origins of even harmonics in grid-connected voltage-source converters, with emphasis on the DC-offset $\rightarrow$ second-harmonic chain, asymmetric switching delays, DC injection with transformer half-cycle saturation, asynchronous low-ratio modulation, and interactions with an unbalanced or pre-distorted grid. The relevant emission and compatibility limits from power-system standards are summarized, and practical mitigation measures are given.

1 Introduction

A three-phase AFE rectifier operated with a symmetric modulation strategy and an ideal power stage draws line currents containing only the characteristic harmonic families of a balanced six-switch converter: the low-frequency characteristic orders $h = 6k \pm 1$ ($k = 1, 2, \dots$) when the converter is current-source-like or line-commutated, and, for a PWM voltage-source converter, sidebands clustered around integer multiples of the switching frequency [1, 2]. Even-order components ($h = 2, 4, 6, \dots$) and a DC component are *non-characteristic*: they are forbidden by the waveform symmetry of the ideal converter.

In practice, small even-order components—most often a second harmonic accompanied by a DC component—are routinely measured on high-power AFE installations. Because they are non-characteristic, they carry diagnostic information: each possible origin corresponds to a specific defect or asymmetry. This note collects these origins in one place and links the converter-level (power electronics) view to the network-level (power systems) view, where even harmonics interact with transformer saturation, protection relaying, and compatibility limits.

2 Half-Wave Symmetry and Non-Characteristic Harmonics

2.1 Fourier condition

A periodic waveform $f(t)$ with period T has *half-wave symmetry* if

$$f\left(t + \tfrac{T}{2}\right) = -f(t). \tag{1}$$

Writing the Fourier series $f(t) = \sum_n \left[a_n \cos n\omega_1 t + b_n \sin n\omega_1 t\right]$ and imposing (1) gives

$$a_n = b_n = 0 \qquad \text{for all even } n \text{ (including } n = 0). \tag{2}$$

Hence a waveform satisfying (1) contains *no* DC component and *no* even harmonics. In a balanced three-phase converter the modulating signals, the switching pattern, and the resulting

line-to-line voltages all satisfy (1) in the ideal case; the line currents inherit the property through the linear grid impedance. Every observed even harmonic is therefore the signature of a violated symmetry: the positive and negative half-cycles are no longer mirror images of each other. DC and even orders are two faces of the same defect and almost always appear together [9, 10].

2.2 Frequency mapping between *abc* and *dq* frames

Many of the mechanisms below are best understood through the frequency mapping of the synchronous (dq) reference frame rotating at the fundamental angular frequency ω_1. A positive-sequence component at frequency ω in the stationary frame appears at $\omega - \omega_1$ in the dq frame; a negative-sequence component at ω appears at $\omega + \omega_1$. The mappings relevant to this note are:

abc (stationary)	sequence	*dq* (synchronous)
DC offset	–	$1\omega_1$ ripple
fundamental	positive	DC
fundamental	negative	$2\omega_1$ ripple
2nd harmonic	negative	$3\omega_1$ ripple
2nd harmonic	positive	$1\omega_1$ ripple
5th (neg.) and 7th (pos.)	–	$6\omega_1$ ripple

Two entries deserve attention. First, a DC offset in the measured *abc* currents and a positive-sequence second harmonic *both* map to a $1\omega_1$ ripple in dq; a synchronous-frame current controller cannot distinguish them and, worse, converts one into the other, as shown in Section 3.1. Second, grid unbalance (negative-sequence fundamental) maps to $2\omega_1$ in dq, which explains the well-known $2\omega_1$ DC-bus ripple under unbalanced conditions [3, 4].

3 Origin Mechanisms

3.1 DC offset in the current (or voltage) measurements

The most common single cause of second-harmonic emission in AFE converters is a DC offset in the current-sensing chain (Hall sensor, burden resistor, analog conditioning, or ADC). The mechanism is a closed-loop chain:

1. A sensor offset adds a spurious DC component to the measured *abc* currents.
2. After the Park transformation, this DC error appears as a $1\omega_1$ ripple superimposed on the dq current feedback.
3. The current controller, whose bandwidth normally exceeds ω_1 by a comfortable margin, reacts to this fictitious ripple and injects a real $1\omega_1$ component into the dq voltage reference.
4. Transformed back to *abc*, a $1\omega_1$ component in dq produces a DC component *and* a second-harmonic component in the actual converter voltages and hence in the line currents.

The chain *offset* $\rightarrow$ *DC + 2nd harmonic* is so repeatable that the simultaneous presence of a DC component and a second harmonic in the line current is practically a diagnostic fingerprint of measurement offset. The analogous mechanism in motor drives, where offset and gain errors of the current sensors produce first- and second-harmonic torque ripple, was analyzed by Chung and Sul [5]; the AFE case is formally identical with the grid replacing the machine. Gain *mismatch* between phase sensors, by contrast, creates a spurious negative-sequence fundamental ($2\omega_1$ in dq) and leads to odd, not even, distortion—the distinction between offset errors (even orders) and gain errors (odd orders) is useful in commissioning.

Voltage-measurement offsets act similarly through the PLL and the feedforward path: an offset on the grid-voltage measurement modulates the PLL angle at $1\omega_1$, which again produces DC and second-harmonic components in the synthesized converter voltage.

3.2 Asymmetric switching delays and dead time

Every switching event of a converter leg introduces a volt-second error with respect to the ideal waveform. With dead time t_d, turn-on delay t_{on}, and turn-off delay t_{off}, the average voltage error of one leg over a switching period T_{sw} has magnitude

$$\Delta V \;=\; \frac{t_d + t_{on} - t_{off}}{T_{sw}}\, V_{dc} \;\cdot\; \mathrm{sign}(i), \tag{3}$$

where the sign follows the leg current polarity. As long as the error is *identical* for both current polarities—i.e., the top and bottom devices and their gate drivers are matched—the distortion in (3) is an odd function of the current and preserves half-wave symmetry: the classical dead-time distortion therefore contains only odd harmonics (predominantly 5th and 7th in a three-phase system, since triplen components cancel) [6, 7].

Even harmonics appear when the error is *different* between the two half-cycles:

- unequal propagation delays of the top and bottom gate-driver channels;
- asymmetric dead-time insertion (different t_d for rising and falling edges);
- parameter spread between the top and bottom devices (t_{on}, t_{off}, $V_{CE,\mathrm{sat}}$ or $R_{DS,\mathrm{on}}$);
- unequal snubber or gate-resistor values after service or rework.

Any such top/bottom asymmetry adds a polarity-independent term to (3), i.e. a DC voltage error per leg, plus even-order distortion. At high power levels with low switching frequency the relative weight of a fixed nanosecond-scale mismatch grows with $1/T_{sw}$ being small, but the volt-second error per event is applied fewer times per fundamental period; in practice, mismatches of tens to hundreds of nanoseconds between driver channels are measurable in the line-current spectrum of megawatt converters.

3.3 DC injection and transformer half-cycle saturation

Whatever its origin, a DC component injected by the converter flows through the line-side magnetics. A transformer offers essentially no impedance to DC other than its winding resistance, so even a small DC voltage bias drives a substantial DC magnetizing current. The flux is displaced from its symmetric trajectory and the core saturates on one half-cycle only. The resulting magnetizing current is a train of unipolar peaks, extremely rich in second-harmonic content—the same phenomenon as transformer inrush, which is precisely why differential protection relays use second-harmonic restraint to distinguish inrush from internal faults [13].

Two aspects make this mechanism particularly relevant:

- **Self-reinforcement.** Half-cycle saturation itself produces even harmonics and DC in the current, which can interact with the converter control and with other transformers in the network, sustaining or amplifying the asymmetry.
- **Regulatory limits on DC injection.** Interconnection standards constrain the DC current a converter may inject: IEEE Std 1547 limits it to 0.5 % of rated current [18]. The limit exists largely because of the saturation mechanism above, including its effect on distribution transformers and on residual-current devices.

The same physics operates at network scale during geomagnetic disturbances: geomagnetically induced currents (GIC) are quasi-DC currents that drive transmission transformers into half-cycle saturation, generating large even (and odd) harmonic currents and reactive-power consumption across entire networks [14]. An AFE installed on a grid affected by transformer saturation—whatever the cause—will see a background second-harmonic voltage and respond to it (Section 3.6).

3.4 Low and asynchronous modulation ratio

At the power levels typical of megawatt AFE converters the switching frequency is low and the frequency-modulation ratio $m_f = f_{sw}/f_1$ may be below ~ 21. In this regime the classical PWM harmonic theory [2] prescribes *synchronous* modulation with m_f an odd integer (ideally an odd multiple of three): the carrier is phase-locked to the fundamental, the pulse pattern repeats identically every fundamental period, and half-wave symmetry of the switched voltage is guaranteed, so no even harmonics and no sub- or interharmonics are produced.

If the modulation is asynchronous (fixed carrier not locked to the grid) with a small non-integer m_f, the pulse pattern drifts with respect to the fundamental. The switched waveform is no longer periodic over one fundamental cycle, half-wave symmetry is lost on a cycle-by-cycle basis, and the spectrum contains even harmonics, subharmonics, and interharmonics whose amplitude grows as m_f decreases. The remedy is structural, not a tuning matter: lock the carrier to the PLL angle and enforce an odd (preferably triplen-odd) m_f, or adopt an optimized synchronous pattern (selective harmonic elimination) that embeds half-wave and quarter-wave symmetry by construction.

3.5 Grid unbalance and control interaction

Grid unbalance is frequently blamed for even harmonics, but the mapping must be kept precise. A negative-sequence fundamental component (unbalance) appears as a $2\omega_1$ ripple in the dq frame and as a $2\omega_1$ power/voltage ripple on the DC bus. If the DC-bus voltage controller is fast enough to chase this ripple, it modulates the d-axis current reference at $2\omega_1$; back in the stationary frame this produces a *third* harmonic (positive sequence) and an additional negative-sequence fundamental—odd components, not even ones [3, 4].

The correct attribution is therefore:

- **Offset-type asymmetries** (sensor offsets, top/bottom leg asymmetry, DC injection) → DC and *even* harmonics;
- **Gain-type asymmetries and unbalance** (sensor gain mismatch, negative-sequence grid voltage, fast DC-bus loop) → negative-sequence fundamental and *odd* harmonics (3rd in particular).

Keeping the two families separate greatly accelerates root-cause analysis on site: the parity of the dominant non-characteristic harmonic identifies the class of defect before any hardware is opened.

3.6 Background distortion of the grid

Finally, the converter may simply be responding to a grid that already carries even-harmonic voltage distortion. Sources on the network side include half-wave-rectifier loads, cycloconverters and other asymmetrically fired line-commutated converters, transformer energization (inrush) events nearby, arc furnaces during unstable arcing, and GIC-driven transformer saturation [9, 11, 12]. An AFE presents a low impedance around the fundamental and its controller bandwidth; a background second-harmonic voltage therefore drives a second-harmonic current through the converter admittance unless the control actively rejects it. In emission assessments

this distinction matters: the measured even-harmonic current at the point of common coupling is the superposition of the converter's own emission and its response to background voltage distortion, and the two must be separated before compliance is judged [16].

4 Standards Perspective

Power-system standards treat even harmonics more strictly than odd ones, precisely because they are non-characteristic and symptomatic of defects:

- **IEEE Std 519** limits even current harmonics to 25 % of the corresponding odd-harmonic limits in each frequency range, and additionally restricts DC injection [15].
- **IEC 61000-3-6** provides planning levels and the methodology for apportioning emission limits to distorting installations connected to MV, HV, and EHV systems, with dedicated (lower) indicative planning levels for even orders [16].
- **EN 50160** specifies the voltage-quality characteristics at the supply terminals of public networks, including individual even-harmonic voltage limits that are markedly lower than those of neighboring odd orders (e.g. 2 % for $h = 2$ and 1 % for $h = 4$, against 5–6 % for the low odd orders) [17].
- **IEEE Std 1547** limits DC injection of distributed resources to 0.5 % of rated output current, addressing the transformer-saturation chain described above [18].

5 Mitigation

The countermeasures follow directly from the mechanisms:

- **Offset auto-calibration.** Measure and store the ADC/sensor offsets at every start-up with the converter gated off, and track slow offset drift online (e.g. by averaging the measured currents over an integer number of fundamental periods during operation).
- **Resonant compensation.** Add a resonant term at $1\omega_1$ in the dq-frame current controller (equivalently, at DC and $2\omega_1$ in the stationary frame). This simultaneously rejects residual measurement offset effects, DC injection, and background second-harmonic voltage. Repetitive control achieves the same result for all harmonic orders at once.
- **Gate-path symmetry.** Match propagation delays of top and bottom driver channels, use symmetric dead-time insertion, and apply dead-time / delay compensation based on current polarity. Verify after any driver or device replacement.
- **Synchronous modulation.** At low m_f, lock the carrier to the grid angle and use an odd (triplen) integer ratio, or an offline-optimized symmetric pattern.
- **DC suppression loop.** For transformer-coupled converters, close a dedicated slow loop on the estimated DC component of the line current (or on the transformer magnetizing asymmetry) and trim the modulator offset accordingly [8].
- **Decoupled sequence control.** Under unbalanced grid conditions, use dual (positive/negative) sequence current control and limit the DC-bus voltage loop bandwidth well below $2\omega_1$, so that unbalance is not converted into additional low-order distortion.

6 Conclusion

Even harmonics in a three-phase AFE are never intrinsic: they always indicate a broken half-wave symmetry. The practically dominant chain is measurement DC offset producing DC plus second harmonic through the synchronous-frame current controller; the remaining mechanisms—top/bottom leg asymmetry, DC injection with transformer half-cycle saturation, asynchronous low-ratio modulation, and background network distortion—each leave a distinguishable signature. Because offset-type defects generate even orders while gain-type defects and unbalance generate odd orders, the parity of the observed non-characteristic harmonics is itself a powerful diagnostic. Standards on both the emission side (IEEE 519, IEC 61000-3-6) and the compatibility side (EN 50160) reflect this by assigning even orders limits far below those of odd orders, and by constraining DC injection directly.

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